Depression in freezing point

Freezing point of a substance is another important physical property like boiling point. Freezing point is defined as "the temperature at which the solid and the liquid states of the substance have the same vapour pressure". At freezing point, the solid and liquid phases of the substance are in equilibrium. For example, the freezing point of water is 0 ˚C. At this temperature the ice and water are in equilibrium. When a nonvolatile solute is added to water at its freezing point, the freezing point of the solution is lowered from 0 ˚C. The lowering of the freezing point of the solvent when a solute is added is called depression in freezing point (ΔT_f).

From the above graph, we infer that the freezing point (T_f^o) is 0 ˚C as the vapour pressure at this temperature is 1 atm (atmospheric pressure). The vapour pressure versus temperature curve for the solution indicates that the freezing point (T_f) is lower than the 0 ˚C. The depression in freezing temperature (ΔT_f) can be expressed as,

$$\Delta T_f = T_f^o - T_f$$

The experimental results show that the depression in freezing point is directly proportional to the molal concentration of the solute particles.

Hence,

$$\Delta T_f \alpha\ m$$

$$\Delta T_f = K_f\ m \qquad \text{.....(9.30)}$$

Here, 'm' = is the molality of the solution

K_f = molal freezing point depression constant or cryoscopic constant.

If m=1 then $\Delta T_f = K_f$

The K_f is equal to the depression in freezing point for 1 molal solution

Table 9.4 Molal freezing point depression constant for some solvents

S. No.	Solvent	Freezing point (K)	K_f (K.kg. mol^{-1})
1.	Water	273.0	1.86
2.	Ethanol	155.7	1.99
3.	Benzene	278.6	5.12
4.	Chloroform	209.6	4.79
5.	Carbon disulphide	164.2	3.83
6.	Ether	156.9	1.79
7.	Cyclohexane	279.5	20.0
8.	Acetic acid	290.0	3.90

Determination of molar mass of solute from depression in freezing point

If the solution is prepared by dissolving w_B g of solute in w_A g of solvent. then depression in freezing point is given by (9.31)

$$\Delta T_f = \frac{K_f \times w_B \times 1000}{M_B \times w_A} \qquad \text{.....(9.31)}$$

molar mass of a solute can be calculated using (9.31)

$$M_B = \frac{K_f \times w_B \times 1000}{\Delta T_f \times w_A} \qquad \text{.....(9.32)}$$

Example Problem - 5

Ethylene glycol ($C_2H_6O_2$) can be used as an antifreeze in the radiator of a car. Calculate the temperature when ice will begin to separate from a mixture with 20 mass percent of glycol in water used in the car radiator. K_f for water = 1.86 K Kg mol^{-1}

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